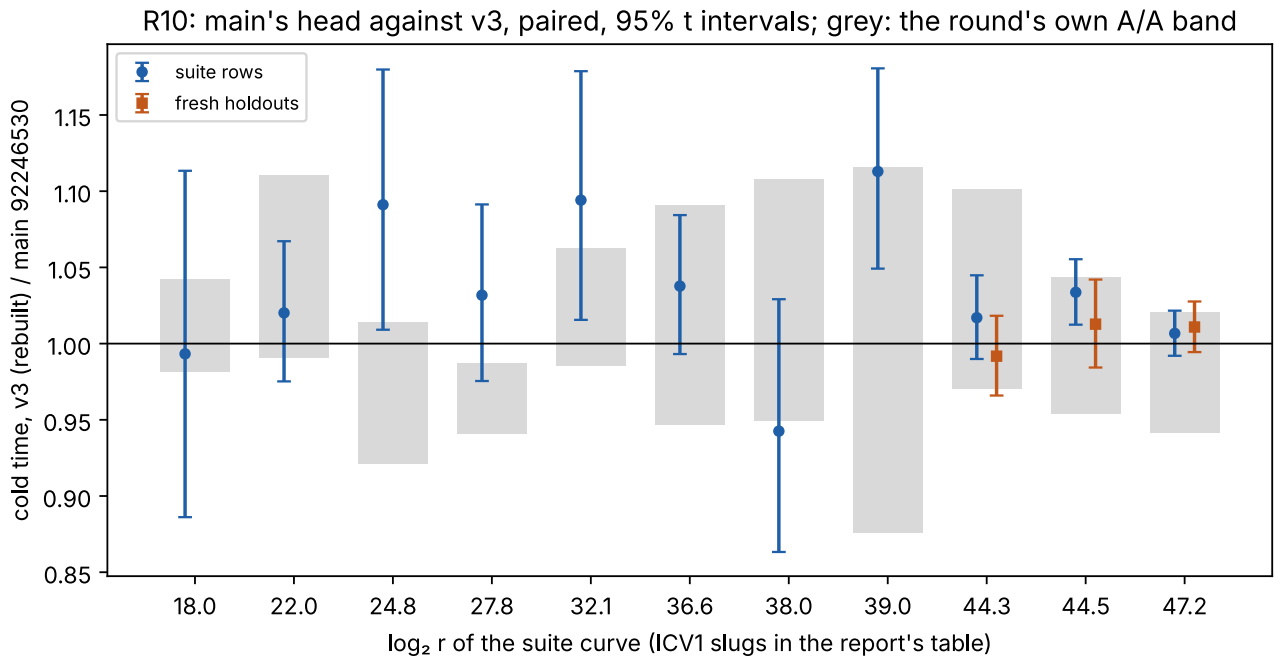
R10: the programme re-based on main's head, again

Accepted, as baseline v4 (engineering). Main at `92246530` against v3 rebuilt from `995ea207`, both built with `rustc 1.97.0`, on an Emerald Rapids host of the reference class (AVX-512 VBMI2, GFNI, VPCLMULQDQ), 2026-10-16. Protocol: `research/ic_tool_program/rounds/R10-main-head/PROTOCOL.md` (PR #1948); analysis: `analysis.json` (`icprog analyse r10`); run trees: `runs.tar.xz` (SHA-256 `5a42268b89d97fca…`).

- **Pins:** both arms equal v0′s outputs on all 90 suite rows.
- **Timed:** 943 isolated processes (A/A 222, suite 400, holdouts 321), 3 contended and re-run, none failed; three container restarts on the same CPU model and kernel build, each resumed and recorded.
- **Decision:** no size regresses beyond the round's own A/A band; callgrind: both arms recover the same logarithm (main 1.26× and 1.45× fewer instructions on the portable paths).



research/ic_tool_program/rounds/R10-main-head/analysis.json; 943 isolated processes, 3 contended (excluded)

Figure 1: Cold time, v3 over main (above 1, main faster), paired 95% t intervals; grey: the round's own A/A band. Values from `analysis.json`.

Suite rows (five rounds)

curve	$\log_2$ r	rows	cold [95%]	S cold, v3 → main	A/A band
`icv1-f2m19-tm797-9c54981b`	18.002	2	0.993 [0.886, 1.113]	122.33 → 117.22	0.981–1.042
`icv1-f2m23-tm5197-1f85e9e1`	22.001	2	1.02 [0.975, 1.067]	31.86 → 30.68	0.991–1.11
`icv1-f2m45-tm6236725-40939294`	24.803	2	1.091 [1.009, 1.18]	20.02 → 18.3	0.921–1.014
`icv1-f2m37-tm534059-32aad96b`	27.781	2	1.032 [0.975, 1.091]	9.05 → 8.59	0.941–0.987
`icv1-f2m43-tm998717-e2e742b0`	32.113	2	1.094 [1.016, 1.179]	5.16 → 4.92	0.985–1.062
`icv1-f2m47-t22705043-f4e44623`	36.636	2	1.038 [0.993, 1.084]	3.46 → 3.32	0.947–1.091
`icv1-f2m57-tm747311035-c1f545af`	38.002	2	0.943 [0.863, 1.029]	3.78 → 3.99	0.95–1.108
`icv1-f2m41-tm2308219-7f48b14a`	39.0	2	1.113 [1.049, 1.181]	4.94 → 4.55	0.876–1.116
`icv1-f2m53-tm56619371-dac20a85`	44.259	8	1.017 [0.99, 1.045]	5.84 → 5.72	0.97–1.101
`icv1-f2m59-tm943548413-98844ecc`	44.517	8	1.034 [1.012, 1.055]	5.48 → 5.18	0.954–1.043
`icv1-f2m61-t158598901-ab42b6c5`	47.211	8	1.007 [0.992, 1.022]	7.71 → 7.65	0.941–1.02

Fresh holdouts (seeds 230–233, T151–T158)

curve	$\log_2$ r	pairs	cold [95%]
`icv1-f2m53-tm56619371-dac20a85`	44.259	40	0.992 [0.966, 1.018]
`icv1-f2m59-tm943548413-98844ecc`	44.517	80	1.013 [0.984, 1.042]
`icv1-f2m61-t158598901-ab42b6c5`	47.211	40	1.011 [0.994, 1.028]

Scope

One host and one core, the reference class. The matched-rho distance at v4 is the rule's comparison's, re-run on v4 later; S at the three target sizes is 5.72, 5.18 and 7.65. R06 is next, ported onto v4 by an amendment before any run.